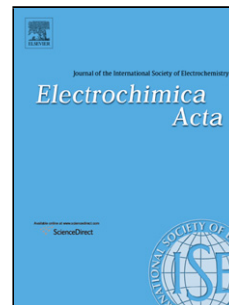


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OXIDE):LiTFSI SYSTEM

Author: K. Pożyczka M. Marzantowicz J.R. Dygas F. Krok



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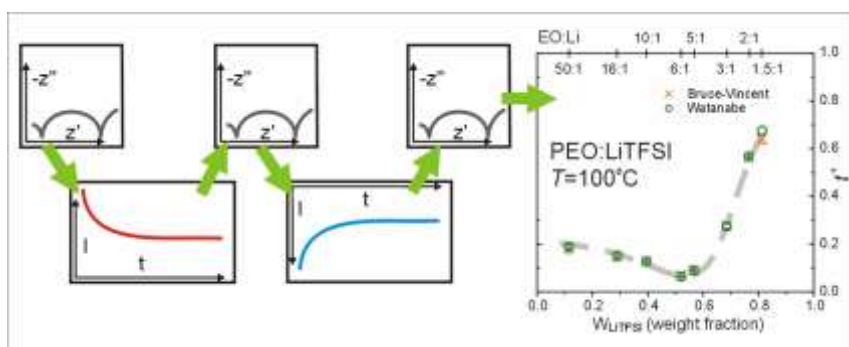
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GRAPHICAL ABSTRACT



Highlights:

New accurate procedure for determination of transference numbers

Lithium transference numbers t^+ studied in a wide range of salt concentration

Low t^+ for salt in polymer and high t^+ for polymer in salt electrolytes

IONIC CONDUCTIVITY AND LITHIUM TRANSFERENCE NUMBER OF POLY(ETHYLENE OXIDE):LiTFSI SYSTEM

K. Pożyczka, M. Marzantowicz, J.R. Dygas, F. Krok*

Faculty of Physics, Warsaw University of Technology, ul. Koszykowa 75
00-662 Warszawa, Poland

** corresponding author, marzan@mech.pw.edu.pl, tel. +48222348213, fax. +48226282171*

Abstract

Polymer electrolytes comprising $\text{LiN}(\text{CF}_3\text{SO}_2)_2$ salt (LiTFSI) and high molecular weight poly(ethylene oxide) PEO are prepared by casting from solution. A wide range of salt concentration is studied, from molar ratio EO:Li of 50:1 to 1.5:1. Samples of electrolytes are measured in symmetrical Li|electrolyte|Li cell. The values of lithium transference number t^+ of electrolytes are determined using a novel procedure, which involves impedance spectroscopy as well as two potentiostatic polarization segments with opposite sign of d.c. bias. The principles of this procedure are based on Bruce-Vincent method, which has been extended in order to compensate the possible differences in properties of the two Li|electrolyte interfaces, and ensure proper conditioning of the cell between different stages of the measurement process. The results are analyzed using Bruce-Vincent and Watanabe approaches, which give similar values of lithium transference number. The reported values of t^+ initially decrease with rising concentration of salt, but after passing a minimum at molar ratio EO:Li of around 6:1, a considerable increase is recorded, eventually leading to lithium transference number higher than 0.5 for “polymer in salt” systems. Such behavior of the studied system is discussed either in terms of formation of PEO:salt complexes, or formation of ionic aggregates.

Keywords: lithium transference number; polymer electrolyte; LiTFSI; poly(ethylene oxide); cation conductivity

1. Introduction

The choice of materials for solid electrolytes applied in rechargeable lithium-ion cells is usually based on the value of their ionic conductivity, electrochemical stability versus electrode materials and convenient mechanical properties. An equally important and often underestimated factor is the ratio of charge transported by lithium ions to the overall charge transport – quantitatively described by the cation transference number t^+ . Unfortunately, for most electrolytes based on polymers with dissolved lithium salts the values of cation transference number are found to be low, often below 0.5. Apart from decrease of efficiency of charge transport process in the cell, low cation transference number can possibly lead to unwanted phenomena, like precipitation of salt at the anode and depletion of electrolyte at cathode [1]. As an example of great practical importance of lithium transference numbers, the goals set by United States Advanced Battery Consortium (USABC) can be quoted: for electrolytes with two or more mobile ion species, the conductivity required for application in a battery is 1×10^{-3} S/cm at room temperature, whereas for single ion conducting electrolyte the threshold is much lower and equals 1×10^{-4} S/cm [2].

Correct determination of lithium transference number t^+ for polymer electrolytes often represents a challenging task. Most of experimental methods proposed so far involve assumptions, which can be fulfilled only in part for polymer electrolytes or involve calculation of quantities which are difficult to measure for solids. Hittorf method [3] and Tubandt method [4], based on measurement of changes of mass or concentration of ionic species in the vicinity of reversible electrodes when passing known amount of charge, are relatively simple and produce accurate results for liquid electrolytes. For polymer electrolytes in the form of thin films, measurement of changes in the concentration of ionic species is very difficult to carry out as the concentration gradients may quickly expand into the entire electrolyte volume after applying an external voltage, or simply because of the limited precision with which the concentration changes at the electrodes can be determined. For liquid electrolytes the ions move together with their solvation shell, and therefore the values obtained by the Hittorf method were also described as the “apparent” transference numbers.

Concentration cell method [5] involves the analysis of a junction potential between two electrolytes composed of the same solvents and salts but with different salt concentration. For this method, however, it is necessary to establish the dependence of both the transference number and the activity coefficient on the salt concentration. Another method - pulsed field gradient nuclear magnetic resonance imaging (pfg-NMR) [6] involves predicting contributions of ionic species to the overall conductivity in respect to the Nernst-Einstein relation between carriers mobility and diffusion coefficients. Condition of total ion dissociation, which is one of the assumptions of this method, may not be fulfilled in many types of solid electrolytes, especially in polymer electrolytes with high salt concentration.

Alternating current method, proposed by Sørensen and Jacobsen [7], is based on comparison of sample impedance values measured at high and very low frequencies. Due to low values of ion diffusion coefficients, measurements of transference numbers in polymer electrolytes by this technique might require very long time, which makes it vulnerable to possible time instability of the measured system. Method described by Bruce and Vincent [8] combines measurements of impedance spectra and data obtained by potentiostatic polarization of sample. The measurements are performed in a symmetrical cell with electrodes reversible for the lithium cation – in practice, lithium metal plates are usually applied. This method allows for relatively quick determination of t^+ but, in the original approach, seems to be vulnerable to numerous error sources, originating not only from measuring equipment but also from preparation and conditioning of the measurement cell. Another method, which involves galvanostatic measurements has been developed by Newman et al [9]. This approach, intended for electrolytes with non-uniform distribution of ion concentration in the sample, has been described as relatively simple and independent from “unwarranted” assumptions. However, also in this case calculation of t^+ involves a number of experimental quantities, each adding a significant uncertainty to the final value.

As the methods described above are based on measurement of different physical quantities, and represent theoretical assumptions which can be fulfilled only partly in case of polymer electrolytes, the values of transference numbers may also show substantial differences. Bruce and Vincent [10] have proposed to divide the methods into three classes, with class I and II methods probing only the transport of

charged species, and class III methods probing diffusion coefficient of all species (charged and neutral) which contain given constituent of the electrolyte. Following this classification, a differentiation between transference numbers (related to fraction of charge carried by cations or anions) and transport numbers can be made. Recalculation of both quantities requires a well-defined model of ionic transport in the system. Such models including the influence of dissociation and formation of ionic pairs on the ionic transport were presented for PEO-based electrolytes [11,12].

For PEO:LiTFSI system, studied in our work, the values of t^+ reported so far also exhibit a strong variation, depending on method of their estimation. The dependence of t^+ upon concentration of salt was studied by Gorecky et al. [13,14] and Edman et al.[15]. In the work of Gorecky et al. [13], based on results of pfg-NMR, values of the lithium transport number obtained for electrolytes with molar ratio between 20:1 and 6:1 EO:Li were generally lower than 0.3 and exhibited a minimum of about 0.16 for electrolyte with EO:Li of 8:1. However, later publication from the same group [14] shows that when the measurements were performed under external electric field, much higher value of cationic contribution to conductivity was obtained ($t^+ = 0.57$ for EO:Li of 8:1). On the other side Edman et al. [15] presented results of electrochemical measurements based on the Newman approach and concentrated solution theory. The obtained values of transference number exhibited a steadily increasing trend with increasing amount of salt, starting from $t^+=0.2$ for EO:Li of 30:1, and eventually reaching 0.6 at EO:Li of 5:1. Numerous publications report transference and transport numbers for just one electrolyte composition. The calculated values of t^+ for electrolytes with EO:Li up to 5:1 fall usually between 0.05 and 0.3 [16-21]. As reported recently, values of t^+ calculated according to diffusion coefficients depend also on the length of PEO chains used as electrolyte matrix, and systematically decrease with increasing molecular weight of PEO [22]. Low values of t^+ have been obtained also for LiTFSI dissolved in polymer matrix based on PEO-MEEGE copolymers with comb-like structure (0.07 for EO:Li=25:1) [23] and with network structure (0.05-0.08 for molar ratio 17:1 EO:Li) [24].

Also the results of the reports on temperature dependence of lithium transference numbers are not unambiguous. The transport numbers obtained from pfg-NMR measurements between 60°C and 100°C have

been described as temperature independent in the studied temperature range [13]. On the other hand, values calculated from electrochemical measurements in the range between 50°C and 90°C were reported to show a significant increase with temperature, from 0.11 to 0.18 [16]. Most reports give value only at a single temperature, usually above the melting point of the electrolyte. Therefore, the transport properties of electrolytes with lower melting point are usually reported at lower temperature [17,19], than corresponding properties of electrolytes with higher melting point [13,15]. Values of t^+ obtained for PEO:LiTFSI system by different research groups are summarized in Figure 1. For clarity of presentation, a differentiation between transport numbers and transference numbers has been made, although it should be noted that for some custom-developed methods such division had an arbitrary character.

As PEO:LiTFSI was initially considered to be one of most promising systems in terms of ionic conductivity and electrochemical stability, the low values of lithium transference numbers described above might seem discouraging. Various efforts have been made to improve the share of lithium cations in the overall charge transport. In one of the approaches to this problem, the poly(ethylene oxide) is replaced with other polymer hosts, which create much weaker coordination bonds with lithium. For polycarbonates with LiTFSI salt, values of t^+ as high as 0.8 were obtained for molecular ratio of monomer to salt of 8:1 [25]. Promising results in terms of both ionic conductivity and cation transference number were obtained also for poly(ϵ -caprolactone) (PCL) with carbonate units incorporated in the polymer backbone [21]. The architecture of PEO matrix can be also modified in order to improve lithium transference numbers. Recent reports show that in system comprising LiTFSI and PEO-based network obtained by UV-induced copolymerization, the transference number of Li^+ increases to 0.55 [26].

Another good way of improvement of cation transference numbers may be anion trapping. High lithium transference numbers (around 0.8) were obtained for electrolytes composed of poly(ethylene glycol) and lithium triflate, trapped in porous structures made of SiO_2 or Al_2O_3 . For such structures, the anions may become trapped on the extensive surface of the porous material [27].

An improvement of lithium transference numbers in PEO-based electrolytes can be achieved also by application of novel lithium salts. The transference numbers t^+ of salts with anions of Hückel-type,

considered as one of the best candidates for practical application in rechargeable batteries, were found to be generally higher than these of PEO:LiTFSI system, ranging from 0.2 to 0.5 [28]. Increase of cation transference number above 0.5 has been also obtained by addition of anion-trapping compounds [29]. Immobilization of anions can be achieved by extending of their structure, which frequently provides also delocalization of charge required for efficient dissociation process. Values of t^+ approaching 0.8 have been reported for PEO-based electrolytes with salts in which the anions contained oligomeric oxyethylene groups [30]. Such oligomeric salts can perform well even in the absence of polymer matrix [31] as they share features of ionic liquids and polyelectrolytes.

As already mentioned above, broad variation of results obtained for the same electrolyte composition indicates that most of methods, which have been applied for estimation of t^+ , are either vulnerable to experimental errors or are based on assumptions unachievable in practice. The equations used for calculation of t^+ are often simplified and do not take into account all physical processes, which take place in the measurement cell. It is also remarkable that the values of transference number were estimated only for electrolytes with low or medium concentration of salt, with the lowest EO:Li molar ratio of 5:1. The increasing trend in t^+ shown by Edman et al. [15] indicates that salt-rich systems and “polymer in salt” electrolytes may be favorable for transport of Li cations. Electrolytes with low salt concentration, regarded so far as the “best choice” for rechargeable cells basing on their high total ionic conductivity, may exhibit poor performance when it comes to lithium conductivity. The main aim of our research were to develop a reliable, easy to perform and relatively quick method of evaluation of lithium transference number, and then to apply this method to polymer electrolytes PEO:LiTFSI, representing a wide range of salt concentration.

2. Experimental

Polymer electrolytes were synthesized by casting from solution. Weighted amounts of high molecular weight PEO (Aldrich, $M_w=5\times10^6$ g/mol) and anhydrous LiTFSI salt (Aldrich) were mixed in acetonitrile solvent in an argon-filled glove box. The solution was poured onto flat PTFE dishes and subjected to a two-stage drying process. The first stage involved drying under low vacuum, which allowed

to form a thin foil. In the second step, which lasted at least one week, the remaining solvent has been removed under high vacuum. The investigated electrolytes represent following molar ratio EO:Li: 50:1, 16:1, 10:1, 6:1, 5:1, 3:1, 2:1 and 1.5:1. Electrolytes with higher amount of salt were not investigated, as according to our previous results [32] they do not form homogeneous foils. The weight fraction of salt in the system was calculated using equation:

$$W_{LiTFSI} = \frac{M_{LiTFSI}}{(M_{LiTFSI} + n \times M_{EO})} \quad (1)$$

All preparations of measurement cell were carried out in an argon-filled glove box. Samples of polymer electrolyte for transference number and conductivity measurements were cut from the films in a form of thin discs, 16.0 ± 0.5 mm in diameter. The thickness of films was in the range from 0.15 to 0.30 mm depending on the composition. Thicker layers were formed by stacking together and heating up two films. Measurements were performed with two types of electrodes: gold-plated stainless steel (blocking) electrodes, and metallic lithium (reversible) electrodes. Two lithium electrodes, 16 mm in diameter, were made by pressing a small slice of lithium rod (Aldrich) onto two stainless steel mounts. Then the outer surfaces of electrodes covered with lithium were pressed against a PTFE tape. As a result passivated layer of lithium adhered to the tape leaving a shiny surface on lithium electrodes. Each Li|electrolyte|Li symmetrical two-electrode cell was assembled by mounting lithium-covered electrodes in a holder equipped with spring loading and distance control mechanism (Fig. 2). Then the polymer electrolyte sample was gently placed between the electrodes, and the minimum distance between electrodes was adjusted, so a good contact could be maintained during the measurement. A piece of metallic lithium was inserted into the holder, acting as indicator and absorber of humidity. Subsequently the cell assembly was sealed and installed on a Peltier thermoelectric cooling and heating device outside the glove box. A thermocouple was located close to the electrolyte.

Impedance spectroscopy measurements with blocking electrodes were performed in a wide range of temperature, from -40°C to 140°C . The temperature cycle comprised multiple heating and cooling runs, as well as heating run performed after rapid cooling of a molten sample. The impedance spectra were measured

using Novocontrol Alpha-N analyzer, within the frequency range from 10 MHz to 10 mHz, and the amplitude of a.c. signal of 50 mV rms.

The transference number measurements were carried out in a range of temperature from 40°C to 110°C with a step of 10°C using a Potentiostat/Galvanostat Autolab PGSTAT 30 under the control of dedicated software. Prior to the first measurement of a cell assembled with a new electrolyte sample, the cells were left for c.a. 8 h at 90°C for equilibration. Then, an initial measurement of impedance spectrum between 50 kHz and 10 mHz was performed, using an a.c. signal of 10 mV or 30 mV rms (for samples with low “bulk” resistance, the amplitude was 10 mV, and for samples with high “bulk” resistance the amplitude was 30mV).

The transference numbers were determined using a symmetric polarization procedure (hereafter denoted as SPP). This method has been developed especially for the purpose of this work as a modification of the Bruce-Vincent method [8]. SPP procedure is performed on a Li|electrolyte|Li cell and combines measurements of impedance spectra with recording of current upon stepwise applied potentiostatic polarization for several hours until reaching stable values.

The SPP procedure comprises two potentiostatic polarization segments with opposite sign of d.c. bias ($+\Delta V$ and $-\Delta V$). The main reasons for application of such a procedure are:

- i. possible differences in actual state of lithium at each of two electrodes, e.g. due to different surface passivation,
- ii. possible differences in adhesion of electrolyte to lithium surface on each electrode,
- iii. possible temperature or concentration gradient in the volume of electrolyte between lower and upper electrode.

The proposed procedure with reversing polarization is intended to improve repeatability of the obtained value of transference number. The transference number is calculated as a mean of values obtained for these two opposite polarizations.

Details of the SPP measuring procedure are shown in a form of a block diagram in Figure 3. At first, the cell is short-circuited allowing only a limited current to flow. Controlled short-circuiting of the cell is aimed

at equilibration of the cell potential, which reduces possible artifacts and leaves the cell in a well-defined initial state for further measurements. Subsequently, the first measurement of impedance spectrum is executed. Obtained results allow for determination of initial resistance of “bulk” electrolyte (R_{b01}) and initial resistance of solid electrolyte interface (SEI) (R_{i01}). Then, a second short-circuiting of cell is performed in order to equilibrate the sample. Next, the first chronoamperometric measurement is performed: an external potential step $+\Delta V$ is applied to the cell (for samples with low “bulk” resistance $\Delta V=10$ mV and for samples with high “bulk” resistance $\Delta V=30$ mV) and the flowing current is recorded as a function of time. The measurement of current is divided into two stages:

- the first stage with fast sampling rate (intervals between measurements ≤ 10 ms) lasting usually 80 seconds. The external potential step $+\Delta V$ is applied with a delay of about 20 seconds in respect to start of this stage in order to properly determine the starting current. Fast sampling rate improves the accuracy of measurement of initial current (I_{01}),
- the second stage, executed immediately after the first stage, with slow sampling rate (intervals between measurements equal 1 s). The measurements are carried out until a stable value of current (I_{S1}) is reached.

Subsequently, the third short-circuiting of cell and the second impedance spectroscopy scan are executed. Analysis of impedance spectrum by equivalent circuit fit allows for determination of stable-state resistance of “bulk” (R_{bS1}) and resistance of SEI (R_{iS1}) for the first potentiostatic polarization. These values are also used as initial resistance of “bulk” (R_{b02}) and resistance of SEI (R_{i02}) for the second potentiostatic polarization. Next, the fourth short-circuiting of cell followed by the second chronoamperometric measurement are performed. Obtained results allow for determination of initial (I_{02}) and stable-state (I_{S2}) current. Finally, the fifth short-circuiting of cell and the third impedance spectroscopy scan are executed. Analysis of the impedance spectrum allows for determination of stable-state resistance of “bulk” (R_{bS2}) and of SEI (R_{iS2}) for the second potentiostatic polarization.

For each potentiostatic polarization segment the value of t^+ can be calculated by means of two approaches. The first one proposed in Bruce-Vincent method [8] is given by formula:

$$t^+ = t_{BV}^+ = \frac{I_s (\Delta V - I_0 R_{i0})}{I_0 (\Delta V - I_s R_{is})} \quad (2)$$

where:

ΔV is a small d.c. bias applied to polarize the sample,

I_0 is an initial value of current upon polarization with d.c. bias,

I_s is a current reached in the steady-state for sample polarized with d.c. bias,

R_{i0} and R_{is} are resistances of solid electrolyte interface (SEI) before and after polarization, respectively.

The second approach uses formula proposed by Watanabe et al. [33] :

$$t^+ = t_w^+ = \frac{R_{bs}}{\left[\frac{\Delta V}{I_s} - R_{is} \right]}, \quad (3)$$

where R_{bs} is the resistances of bulk after d.c. polarization, and other quantities have the same meaning as those explained in Equation 2.

Lithium cation conductivity was calculated according to formula:

$$\sigma^+ = \sigma_r \cdot t^+ \quad (4)$$

where σ_r stands for ionic conductivity of electrolyte, measured by means of impedance spectroscopy in the cell with lithium reversible electrodes, and t^+ is the cation transference number.

All uncertainties were calculated according to instructions published by Joint Committee for Guides in Metrology in “Guide to the Expression of Uncertainty in Measurement” [34].

3. Results

Exemplary data sets gathered from chronoamperometric and impedance spectroscopy measurements are shown in Figure 4. As described above, a d.c. bias of $\Delta V = 10$ mV or $\Delta V = 30$ mV was applied to the cell and a current flowing through cell has been monitored as a function of time until a steady-state was reached. After sufficient amount of time a steady-state was attained, which in ideal case would correspond to setting up of a uniform salt concentration gradient in the direction of electric field (perpendicular to the surface of electrodes). Numbers 1., 2., 3. in the area of current vs. time plots indicate schematically moments

in time, when impedance spectra were collected (respectively the first, the second and the third impedance spectroscopy scan). Nyquist plots of impedance spectra are shown in the last column in Figure 4.

Cation transference numbers t^+ at the temperature of 100°C calculated for all studied PEO:LiTFSI electrolytes are presented in Figure 5. At this temperature, the electrolytes representing molar ratio ranging from pure PEO to EO:Li of 3:1 are all in amorphous state, which allows for a direct comparison of their properties. Violet crosses mark results calculated according to Bruce-Vincent approach (Eq. 2), and dark green circles correspond to results obtained by means of Watanabe approach (Eq. 3). The obtained results show that the values of transference number do not depend much on the method of calculation. With increasing amount of salt in electrolyte, values of transference number at first decrease, reaching a local minimum for systems with medium salt concentrations. For electrolytes containing more than 50% weight of LiTFSI, a steep increase of the cation transference number with increasing amount of salt is observed, reaching values as high as 0.65 for electrolytes representing the “polymer in salt” regime.

In Figure 6, we have presented transference numbers evaluated for data measured at 70°C. As the temperature was closer to the temperatures used in studies performed previously by other authors, the plot was overlayed onto selected results presented in Fig. 1. The trend observed in our data is similar as described above, and shows a minimum at molecular ratio EO:Li of 6:1. For electrolytes with highest concentration of salt, the calculated values even surpass 0.8, although the uncertainty of those values also increases. This effect is related to relatively higher crystallinity of those samples, which introduces stronger dependence of conductivity upon thermal history of the sample, and other artifacts which affect the process of analysis of the electrochemical data.

The values of “bulk” ionic conductivity of the electrolytes were obtained by analysis of the impedance spectra using equivalent circuit method. As shown in Figure 7. for selected PEO:LiTFSI electrolytes measured at 100°C, similar results are obtained for measurements performed between reversible (metallic lithium) and blocking (gold-platted stainless-steel) electrodes for all studied PEO:LiTFSI electrolytes (the data correspond to conductivity measured at 100°C). This confirms correct assignment of resistances to phenomena observed in the impedance spectra measured between lithium electrodes. A most

significant deviation between ionic conductivity measured in blocking and reversible cells was observed for electrolyte with molar ratio of 3:1 EO:Li. The differences between values of ionic conductivity measured between reversible and blocking electrodes may be caused by both experimental errors (with sample thickness being the most significant source of uncertainty) and by different properties of both types of interfaces. In our previous publication [35] we have shown, that even for simple blocking electrode, the properties of the interface strongly influence the calculated value of resistance related to the “bulk” conductivity. Electrolytes with low concentration of salt, for which the conductivity measured between reversible electrodes is higher, had relatively lower viscosity, and were difficult to separate from the lithium electrodes after measurements. Electrolytes with higher concentration of salt maintained their shape even under spring loading used in the experimental cell, and were also easier to separate from the lithium electrodes.

The trend observed in conductivity values presented in Figure 7 seems to be opposite to that observed in Figure 5 for transference numbers – the electrolytes with high conductivity are characterized by low transference number. In order to compare the influence of these two effects on transport of cations, it seems convenient to calculate the “reduced” cation conductivity (Eq. 4). The values of ionic conductivity measured between metallic lithium electrodes were multiplied by corresponding transference number calculated according to Bruce-Vincent (Eq. 2) or Watanabe (Eq. 3) approach. The obtained values, plotted versus weight fraction of LiTFSI salt were added to the data presented in Figure 7. In the range of salt concentration corresponding to molar ratio EO:Li between 50:1 and 3:1, the cation conductivity of all electrolytes is comparable, with a shallow minimum for EO:Li of 6:1. Significantly lower values were obtained for electrolytes with molar ratio EO:Li of 2:1 and 1.5:1. It should be noted that, according to earlier studies, at 100°C these electrolytes may still contain considerable amount of poorly conductive crystalline phase [32]. Measurements performed between blocking electrodes indicate, that at 100°C presence of this phase affects mainly bulk properties, while effects related to deterioration of polymer/electrode interface (increasing stiffness, decrease of contact area) take place at lower temperature, around 70°C.

For some of studied electrolytes it has been found that the calculated lithium transference number exhibit temperature dependence. As can be seen in Figure 8 and in Table 1, the character of the temperature dependence is different for systems with high and low amount of salt. For polymer electrolytes with the high content of salt the values of t^+ show systematic decrease with increasing temperature. It is remarkable, that the values of lithium transference numbers measured at 70°C for electrolyte representing molar ratio EO:Li of 2:1 are even close to 0.8. On the other hand, the values of t^+ calculated for electrolytes with low and moderate content of salt are low in the whole studied temperature range and do not exhibit distinctive and systematic changes with temperature.

4. Discussion

From the plot of lithium transference number versus the weight percentage of LiTFSI salt it can be concluded, that in electrolytes with low amount of salt, the lithium ions play only a minor role in overall charge transport. Such effect can be explained by different mechanisms of ion motion in the polymer matrix for different salt concentration regimes. For diluted electrolytes, the Li^+ cations are attached to the polymer chain by coordination bonds, and movement along the chain requires a sequence of changes of chain conformation. Moreover, the transport is constrained by the requirement of hopping between different chains. For movement of anions, only free volume is required, which makes it more effective. Increasing freedom of movement of polymer chains with increasing temperature is beneficial for transport of both anions and cations.

For electrolytes representing middle range of salt concentration, e. g. system with EO:Li ratio equal 6:1, the lithium transference number is found to be exceptionally low. In order to explain this effect it should be taken into account, that in this electrolyte the salt forms stoichiometric complex with molar ratio EO:Li of 6:1. It was suggested [36,37] that polymer electrolytes partially preserve local order of crystalline structures even in amorphous state. The exact structure of the $\text{PEO}_6\text{:LiTFSI}$ crystalline complex is not known; however, there are some indications that this structure should be similar to that observed in $\text{PEO}_6\text{:LiAsF}_6$, $\text{PEO}_6\text{:LiPF}_4$, and $\text{PEO}_6\text{:LiSbF}_6$ complexes [38,39]. In such a structure, the lithium ions are supposed to travel

through a “tunnel” created by two PEO chains in helical conformation, whereas the anions become completely separated from cations and move outside such the tunnels. Molecular dynamics simulations show that also in this case transport of anions is much more effective than that of cations [40] and that difficult inter-chain hopping between ends of the tunnels may impose limits on lithium ion conductivity.

Another alternative interpretation of observed dependence of values of transference number on concentration of salt may come from formation of ionic aggregates – pairs and triplets. Earlier studies on PEO:LiTFSI system have confirmed, that in a certain composition range, the portion of such aggregates may increase considerably with salt content. Edman [41] has reported that for the molar ratio EO:Li of 6:1, almost a quarter of mobile species are ionic pairs. With increasing concentration of salt, triplets comprising two anions and one lithium cation can participate in the charge transport. Such negatively charged triplets give negative contribution to the lithium cation transference number [42]. Presence of ionic aggregates may also cause strong deviation of data presented in Fig. 6 from series of measurements performed by Edman et al. [15] using the Newman method. While our method probes directly the lithium transference numbers, in the Newman approach it is obtained by subtraction of anionic transference number.

As the weight fraction of salt rises above 50%, the values of t^+ measured in our studies begin to increase. At first look, this effect seems to be in contradiction to the expected increasing role of ion-ion interactions and formation of ionic aggregates, which are commonly believed to cause suppression of lithium transport. In case of the studied system, ion-ion interactions may actually support transport of lithium ions. Molecular dynamics studies show that ion-ion interactions promote lithium ions hopping between chains [43], which becomes an increasingly important stage of cation transport in systems with high amount of salt. Another explanation for increase of cation transference number, occurring simultaneously with decrease of total ionic conductivity, may come from the specific structure of PEO₃:LiTFSI complex. The model published by Andreev et al. [44] shows that the PEO₃:LiTFSI complex contains one Li cation per each turn of the PEO helix, which is coordinated by three oxygen atoms from polymer, and two oxygen atoms from TFSI⁻ anions. The model indicates also “bridging” of two cations by oxygen atoms from one of the SO₂CF₃ groups of the TFSI⁻ anion. Molecular dynamics simulation indicate, that also in the case of

complexes with EO:Li of 3:1, movement of lithium ions between available positions in the structure is limited by high energy barriers, and therefore depends strongly on relaxations of polymer chain, which decrease the hopping energy [45]. The model indicates that most of the anions may become involved in structure of the complex and no longer move freely in volume between the chains. Large concentration of anions creates also a situation unfavorable for anion transport, in which most of conduction sites are already occupied. Therefore, the decrease of anion transference number, and also decrease of total conductivity of electrolyte may be related to restriction of anion movement. Our studies have shown, that the cation conductivity of electrolyte with molar ratio EO:Li of 3:1 is similar to that of dilute electrolytes (Fig. 7). To some extent, this effect can also be responsible for differences between results obtained using methods based on Bruce-Vincent and Newman approaches: presence of stoichiometric complexes can strongly affect measurements related to concentration of salt, as the concentration of “free” anions and cations may depend rather on phase diagram than total concentration of the salt in the electrolyte.

Alternatively, the observed dependence of cation transference number on concentration of salt can be related to the presence of ionic aggregates. It could be assumed that in electrolytes with high concentration of salt, charged ion aggregates (e.g. triplets such as Li_2TFSI^+ , $\text{Li}(\text{TFSI})_2^-$ or even quadruplets such as $\text{Li}_3(\text{TFSI})_2^+$, $\text{Li}_2(\text{TFSI})_3^-$), may play a significant role in total charge flow because of their possible abundance. Such aggregates with “bound” lithium ions may cause a decrease of ionic conductivity in respect to dilute electrolytes with fully dissociated ions. This can be attributed to the high molecular weight and relatively big volume of ion aggregates. However, these properties of ionic aggregates may have weaker influence on the value of transference number calculated by the presented electrochemical method. For such aggregates the molecular weights of positively and negatively charged species are almost equal (e.g. $M_w(\text{Li}_3(\text{TFSI})_2^+) \approx M_w(\text{Li}(\text{TFSI})_2^-)$), so almost equal electrical force is needed to transport them through the electrolyte.

An interesting and unexpected behavior is observed for electrolytes with high amount of salt (“polymer in salt” systems), in present studies represented by molar ratios EO:Li of 2:1 and 1.5:1. These systems, which usually represent a metastable state caused by proximity to the solubility limit of salt in

polymer [33, 46], in the present study were found to exhibit high values of lithium transference number, significantly surpassing 0.5, and rising steadily with increasing amount of salt. In order to explain this effect, several factors may be considered. First of all, in the “polymer in salt” systems existence of small clusters of amorphous salt is expected, with polymer chains acting like inhibitor of crystallization of those clusters. Inside such clusters, the small size and high mobility of lithium cations give it a significant advantage over bulky TFSI⁻ anions. According to molecular simulation results, transport of Li⁺ inside the clusters occurs via exchange between subsequent anions [47]. As indicated in our recent publication concerning AN-co-BuA copolymer with high amount of LiTFSI salt [48], a situation in which the Li⁺ cation is coordinated by one oxygen from the polymer chain can be regarded as most favorable for ionic conductivity. Such a condition is met along the borders of salt clusters and polymer chains. Higher number of coordination bonds between Li⁺ and PEO would cause either crosslinking of chains, or wrapping of the chain around the cation. The first effect would limit mobility of the matrix, and result not only in decrease of the amplitude of segmental motions, but also possibly in segregation of polymer and salt (as the polymer becomes less effective as a solid solvent). Wrapping of the chain around the cation forces it toward the regime of motion along the polymer chain, which seems to be less-effective in the discussed case. It can be concluded that for the “polymer in salt” systems, an optimum salt concentration should be found, which would compromise high lithium transference number (which increase with amount of salt) and stability (which decreases with increasing amount of salt, as solubility limit of salt in polymer is surpassed).

It is arguable, why the transference numbers decrease with the temperature, as shown for electrolyte with molar ratio EO:Li of 2:1 (Fig. 8). To explain this effect we have to take into account that the increase of temperature supports both transport of anions and cations, but is more beneficial for the ionic species with lower transference number (as it partially removes constraints which block the transport). However, another interpretation can be proposed: with increase of solubility of salt in polymer, more lithium cations come in interaction with the polymer chains, and create more than one coordination bond with the chain. Therefore, the role of ion-chain interaction increases, and the role of ion-ion interaction decreases. As a consequence, the mechanism of their motion switches from motion along clusters of salt to less effective motion along

polymer chains. Another explanation may come from dissociation of ionic clusters into smaller aggregates or free ions. The magnitude of such a process should also increase with increasing temperature.

5. Conclusions

The values of lithium transference number of polymer electrolytes representing PEO:LiTFSI system were estimated by a new symmetrical polarization procedure. It has been found that the values obtained for electrolytes with low and moderate amount of salt (molar ratio EO:Li between 50:1 and 5:1) are all situated below 0.2. The lowest value of t^+ (around 0.07) was obtained for electrolyte of molar ratio EO:Li of 6:1. A significant increase of lithium transference number is observed for salt-rich systems; electrolytes representing “polymer in salt” regime exhibit values of t^+ above 0.5. Such behavior can be interpreted either in terms of trapping of anions in the structure of PEO:salt complexes and clusters of amorphous salt, or alternatively by creation of ionic aggregates, which also contribute to cation conductivity.

Although presented results seem to indicate a possibility to apply “polymer in salt” systems in a lithium rechargeable cell, further studies on conduction mechanism and in particular on transference numbers in such systems are required.

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6. References:

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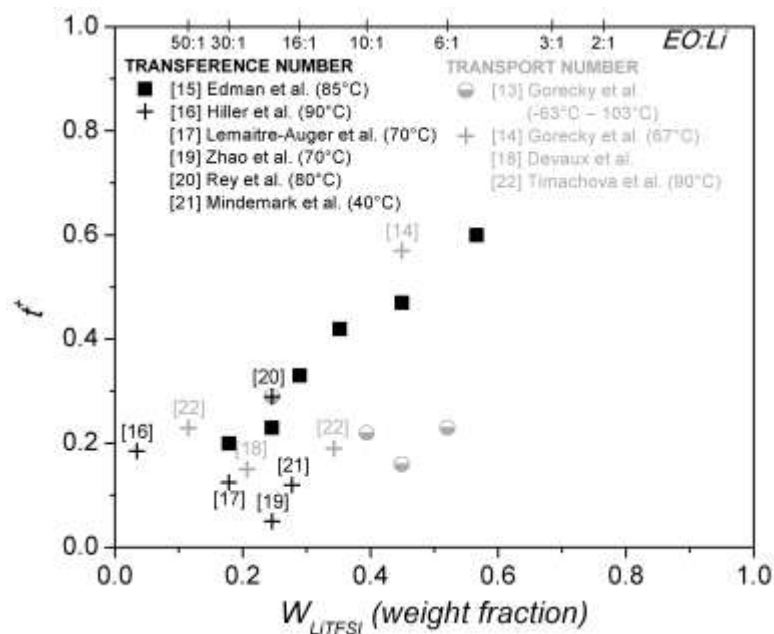


Fig. 1 Values of the cation transference number and lithium transport number obtained for PEO:LITFSI system by different research groups.

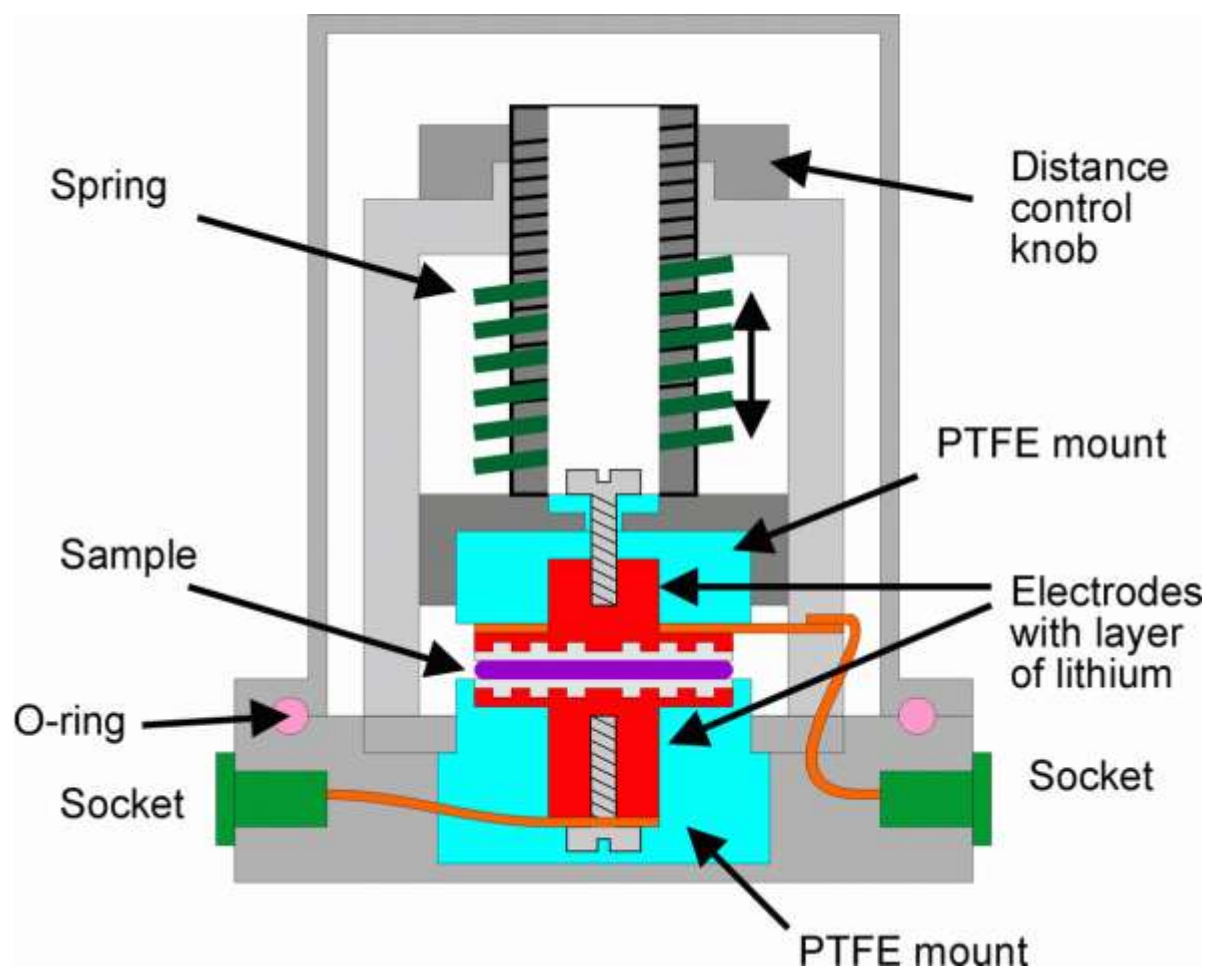


Fig. 2. Schematic presentation of gas-tight sample holder with adjustable spring loading of upper electrode.

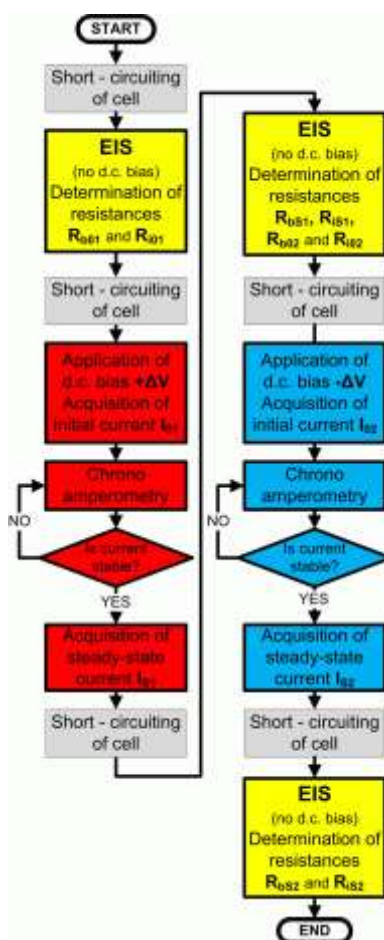


Fig. 3. Block diagram of symmetric polarization procedure for evaluation of Li transference number.

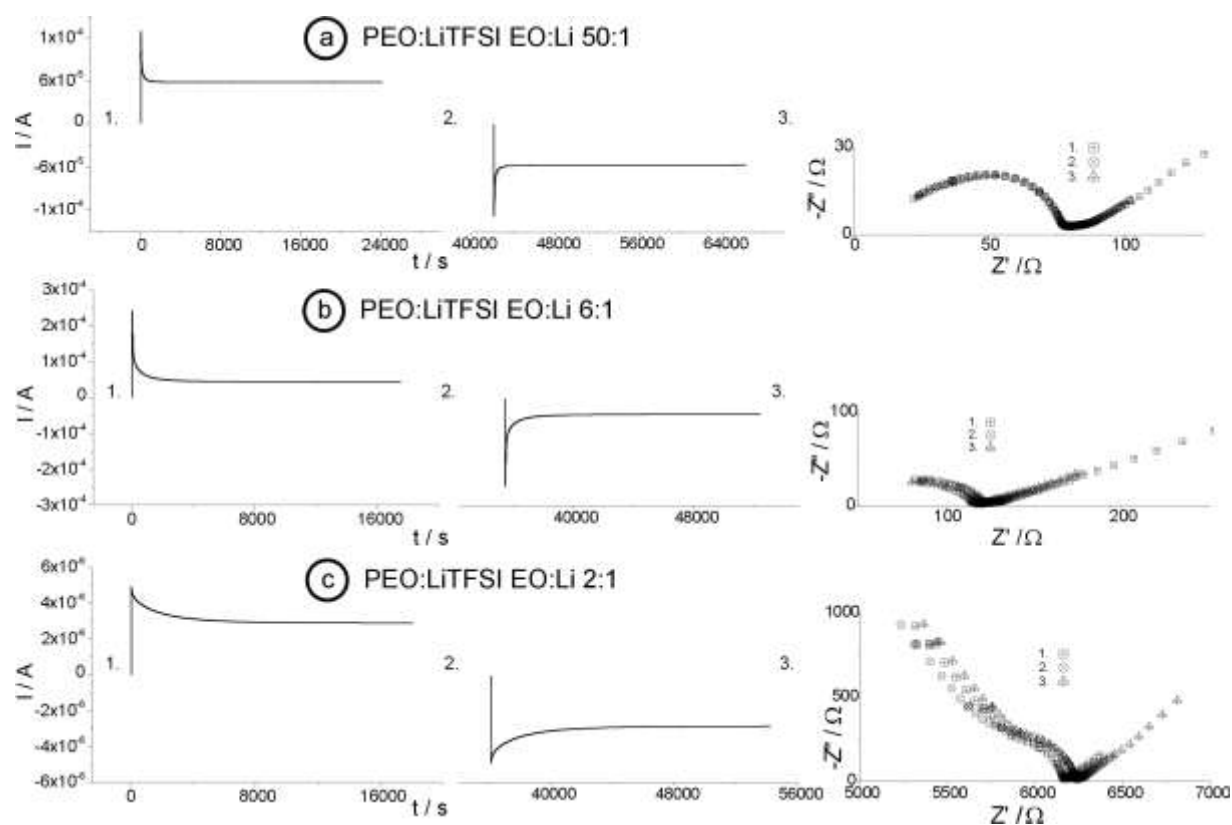


Fig. 4. Exemplary data sets obtained by symmetric polarization procedure at 100°C for PEO:LiTFSI with molar ratio EO:Li of: a) 50:1 b) 6:1 c) 2:1

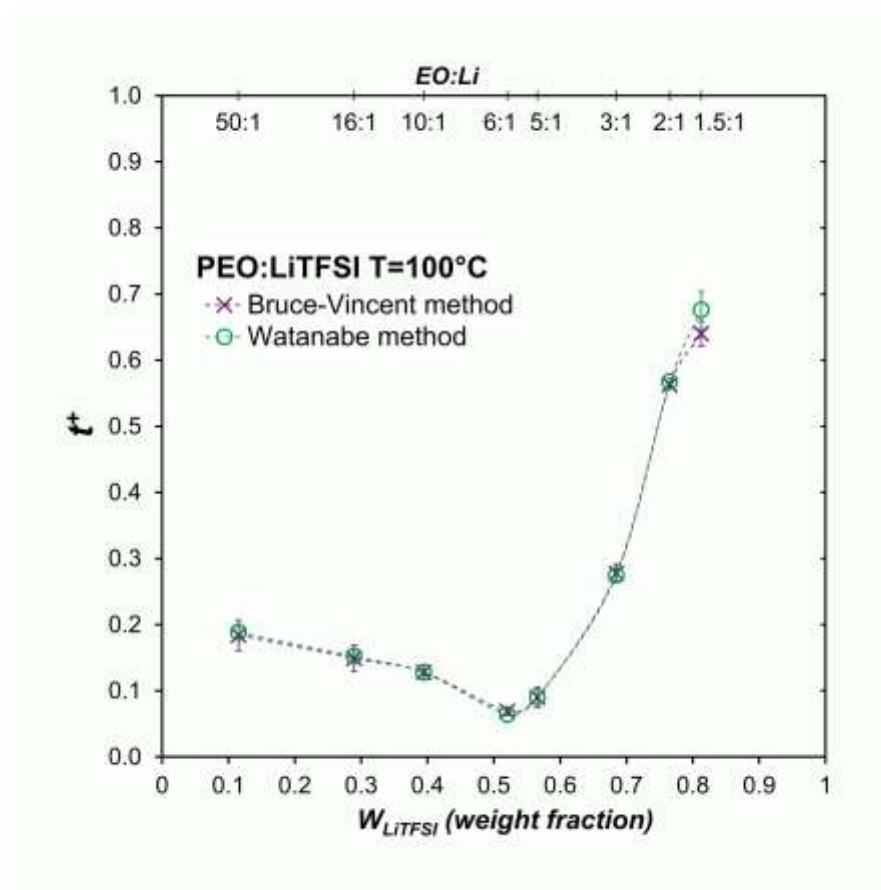


Fig. 5. Values of lithium transference number of PEO:LiTFSI electrolytes obtained at 100°C. The dashed lines are guidelines for the eye. Error bars represent measurement uncertainty.

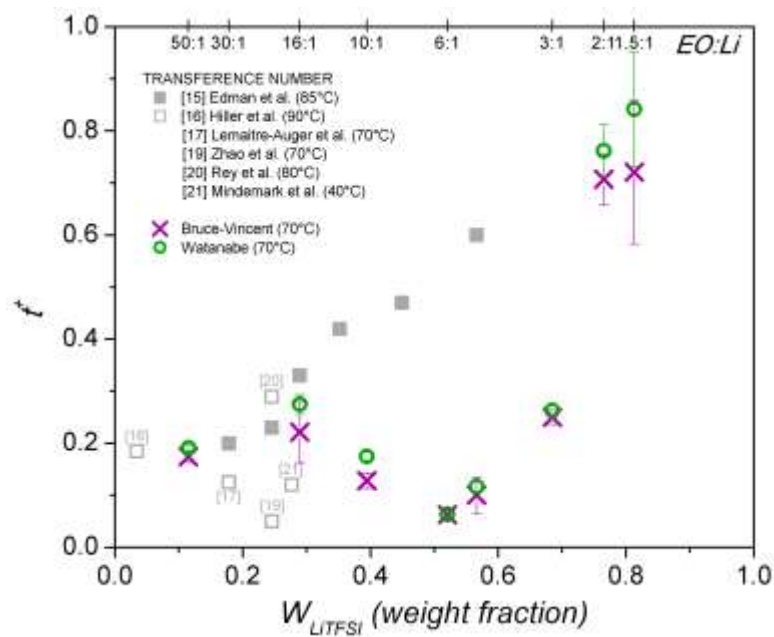


Fig. 6. Values of lithium transference number of PEO:LiTFSI electrolytes obtained at 70°C. Error bars represent measurement uncertainty. Selected reference data from Fig. 1 were added for comparison.

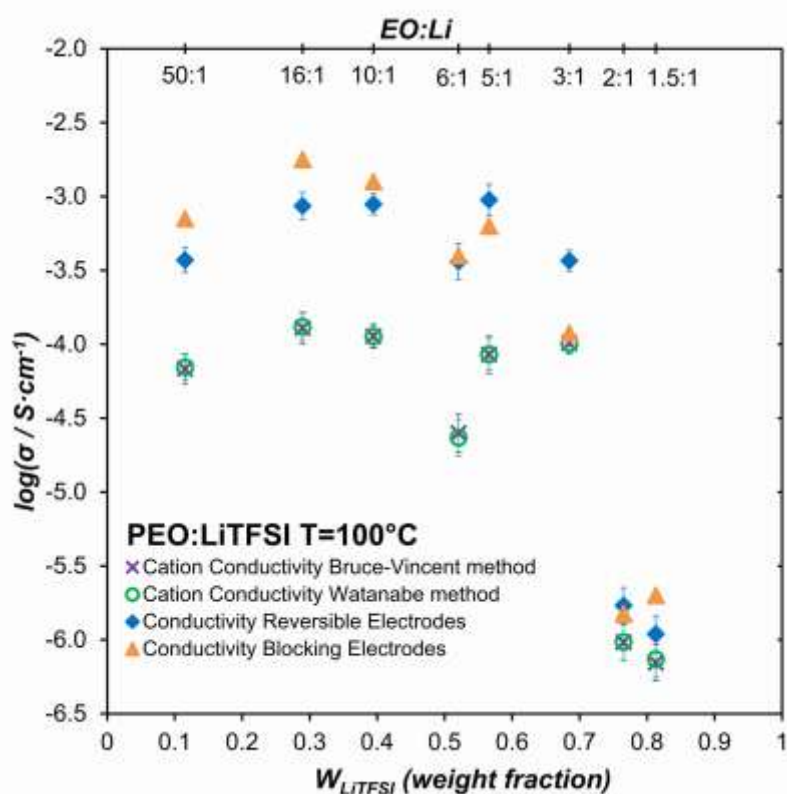


Fig. 7. Ionic conductivity of PEO:LiTFSI electrolytes measured between reversible or blocking electrodes at 100°C , and cation conductivity of PEO:LiTFSI electrolytes at 100°C . Electrolytes with molar ratio EO:Li 2:1 and 1.5:1 are semicrystalline at that temperature.

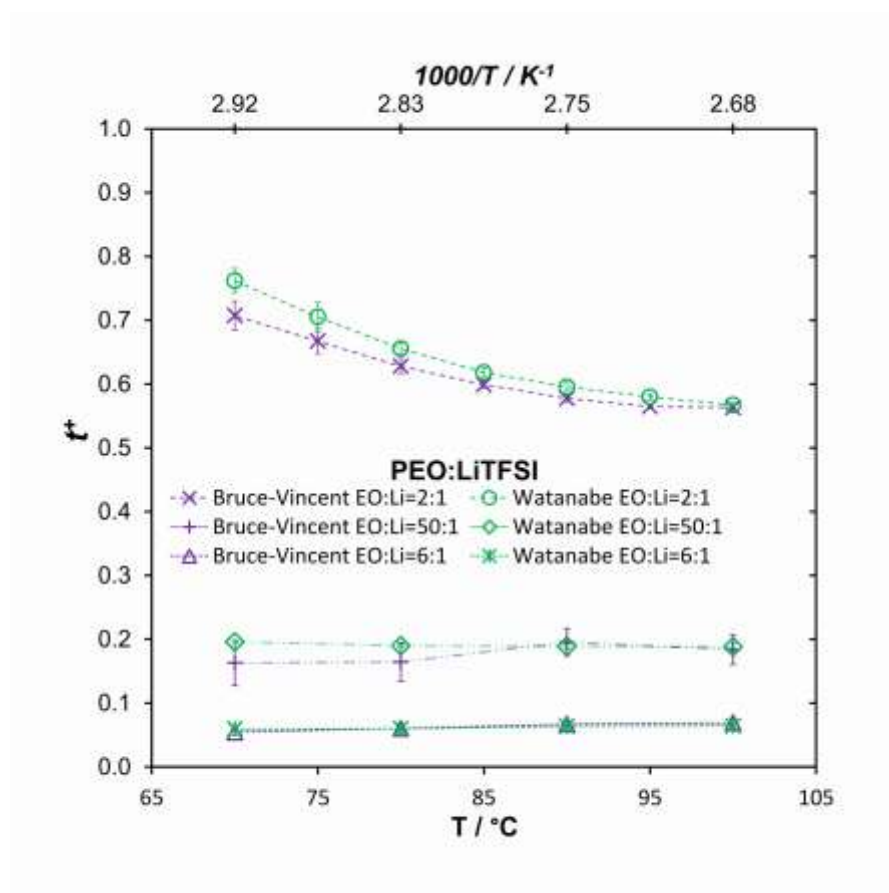


Fig. 8. Temperature dependence of values of lithium transference number for PEO:LiTFSI electrolytes with molar ratio EO:Li of 2:1, 50:1 and 6:1. Vertical bars indicate calculated experimental error.

W_{LiTFSI}	EO:Li	T [°C]	1000/T [1/K]	t^+	
				Bruce-Vincent	Watanabe
0.765	2:1	70	2.92	0.707 ±0.023	0.762 ±0.019
		75	2.87	0.667 ±0.019	0.705 ±0.023
		80	2.83	0.628 ±0.011	0.655 ±0.009
		85	2.79	0.599 ±0.005	0.618 ±0.007
		90	2.75	0.578 ±0.005	0.595 ±0.006
		95	2.72	0.565 ±0.003	0.580 ±0.005
		100	2.68	0.563 ±0.004	0.567 ±0.005
0.521	6:1	70	2.92	0.054 ±0.005	0.059 ±0.001
		80	2.83	0.060 ±0.008	0.060 ±0.001
		90	2.75	0.067 ±0.005	0.063 ±0.001
		100	2.68	0.069 ±0.007	0.064 ±0.002
0.115	50:1	70	2.92	0.163 ±0.036	0.196 ±0.003
		80	2.83	0.164 ±0.030	0.190 ±0.002
		90	2.75	0.195 ±0.022	0.189 ±0.004
		100	2.68	0.184 ±0.023	0.188 ±0.006

Tab. 1. Temperature dependence of values of lithium transference number for PEO:LiTFSI electrolytes with molar ratio EO:Li of 2:1, 50:1 and 6:1.